

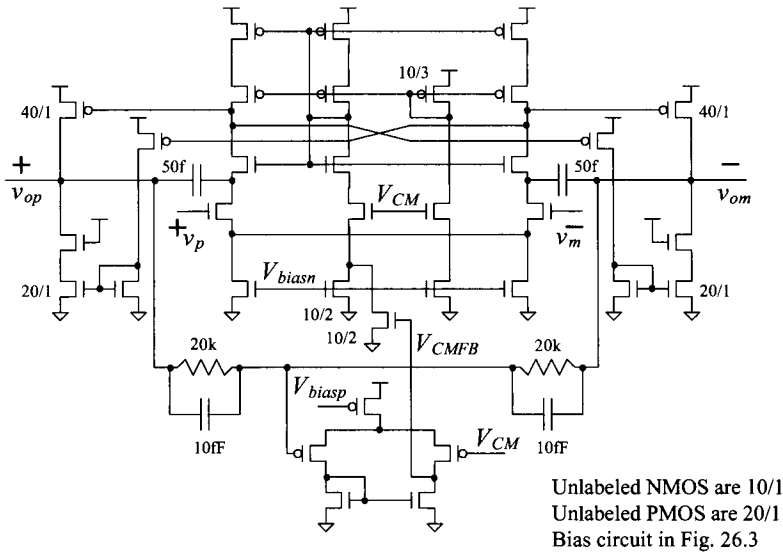


Figure 26.40 Making the op-amp more practical.

A more practical problem is the change in the MOSFET's drain currents with drain-source voltages. For example, looking at Figs. 26.18a and b, we see that it's possible for the NMOS's drain current to be twice the PMOS's drain current for the same gate-source voltages. This can cause op-amp failure in the CMFB circuit. If, for example, the current flowing in the 10/2 MOSFET connected to V_{biasn} is larger than the current sourced by the PMOS, the voltage V_{CMFB} goes to zero and the CMFB loop doesn't work properly. A good "rule-of-thumb" is for the fixed current flowing in the CMFB-controlled bias circuit to be 25–50% of the total expected current.

Increasing the Op-Amp's Open-Loop Gain

The op-amp that we've developed in this section has an open-loop gain in the hundreds. As we'll see in Ch. 29 (Eq. [29.59]), the open-loop gain of the op-amps used in a data converter has a direct effect on the maximum attainable resolution. Towards increasing the gain of the op-amp, let's use the gain-enhancement (GE) techniques presented in Sec. 24.4.

Figure 26.41 shows how we should *not* implement GE. An amplifier with fully-differential outputs, like the amplifier in Fig. 26.7, regulates the drains of the top PMOS devices. We know that this is bad because we would need a CMFB circuit to balance the outputs of the added amplifier.

Figure 26.42 shows how we can add GE to the op-amp developed in this section. Notice that GE is implemented with amplifiers having single-ended outputs. When designing the added amplifiers, as discussed in Sec. 24.4, the bandwidth isn't important for high-speed operation. The lengths of the MOSFET in the added amplifiers can be increased to reduce power and boost gain. An important concern is the added amplifier's input common-mode range.

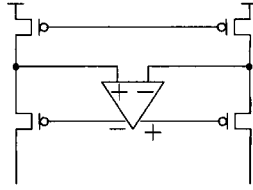


Figure 26.41 How not to implement GE in an op-amp (unless the added amplifier employs CMFB).

The practical problem with the topology seen in Fig. 26.42 is the implementation of the CMFB. With the GE added to the op-amp we now have four additional feedback loops. Variations in V_{CMFB} affect all GE loops in addition to the CMFB through the op-amp. Making these loops stable becomes extremely challenging. What we need is to implement the CMFB without including the diff-amp and GE amplifiers. We can only do this by controlling the output buffer common-mode level, Fig. 26.43. This circuit includes the output buffer we used in Fig. 26.37. Now, however, we add an amplifier in series with the triode-operating MOSFETs to boost the CMFB gain (so that the outputs can be balanced around V_{CM}). The problem with this approach is that the CMFB isn't compensated using the same capacitors as the differential forward signal path (i.e., no Miller effect). We must be concerned with CMFB stability.

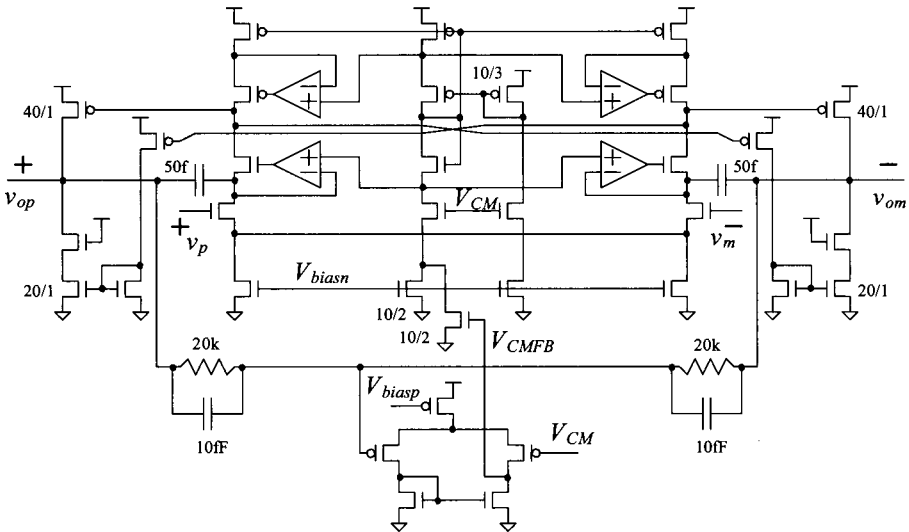


Figure 26.42 Adding gain-enhancement to the op-amp.

Let's discuss the op-amp design in Fig. 26.43. We begin by showing the DC behavior of the op-amp (to show that the outputs are indeed balanced). Figures 26.44a and (b) show the DC behavior and gain of the op-amp in Fig. 26.43 in the topology seen in Fig. 26.24. Again note that by increasing the values of the 20k resistors used to average

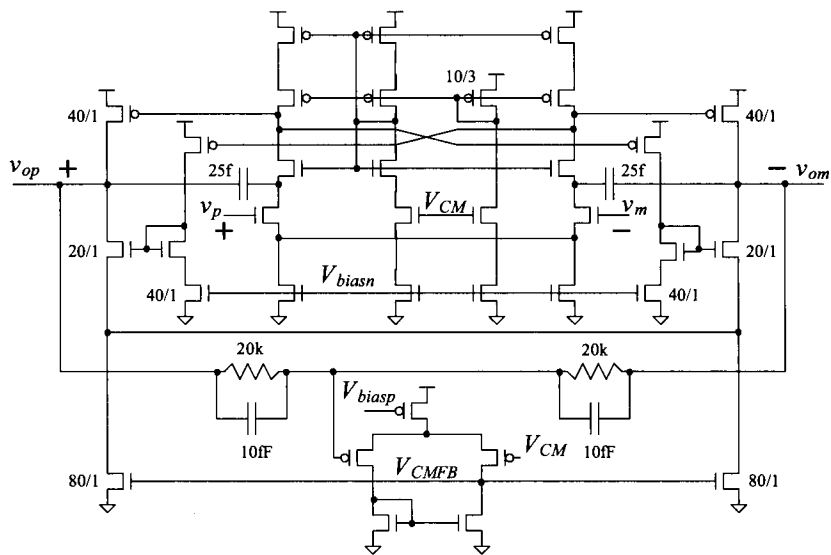


Figure 26.43 Providing CMFB through just the output buffer. Using an amplifier with triode-operating MOSFETs for CMFB (good).

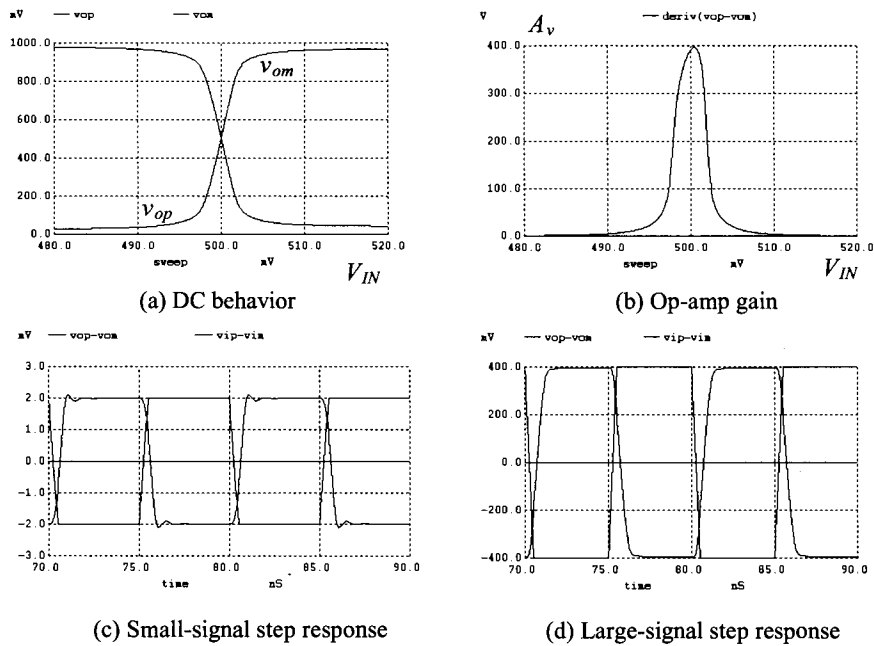


Figure 26.44 Behavior of the op-amp in Fig. 26.43 (see text).

the outputs we can increase the DC gain of the op-amp. Because the forward differential path compensation (now) doesn't include the CMFB path, we've reduced the compensation capacitors from 50 fF to 25 fF (there are practical issues with this change that we'll discuss next). Note that the widths of the added triode-operating MOSFETs are increased to ensure that they operate well within the triode region and don't significantly affect the output drive capability of the op-amp. Figures 26.44c and (d) show the small- and large-signal step responses (see Fig. 26.24). The settling times are approximately 1–2 ns. The current drawn from V_{DD} (including the the bias circuit current) is approximately 250 μA (200 μA for the op-amp alone).

Offsets

We said in Sec. 26.2 that we want to add offsets of 50 mV in series with the gates of our MOSFETs to see if the op-amp “breaks.” Consider the addition of an input-referred offset voltage in the schematic seen in Fig. 26.45. This offset is used to model the overall offset voltage of the op-amp (see Fig. 24.4). If we look at the effects of the offset on the circuitry in Fig. 26.43, we see that one of the NMOS devices in the diff-pair will have a higher overdrive voltage (larger g_m) than the other NMOS device. Since the unity-gain frequency of an op-amp is given by $g_m/2\pi C_c$, the effect is a shift in the unity-gain frequency (and potential instability). As seen in Fig. 26.45, using the 25 fF compensation capacitors destabilizes the op-amp (as indicated by the ringing). Increasing the compensation capacitor's value (back) to 50 fF makes the step response cleaner. We could argue that 50 mV is an unrealistically high offset voltage, so it's better to leave the capacitors at 25 fF (and this may be the case). However, in a practical CMOS process, the op-amp's characteristics shift with the process shifts (and temperature). It's better to overcompensate than to have an unstable op-amp.

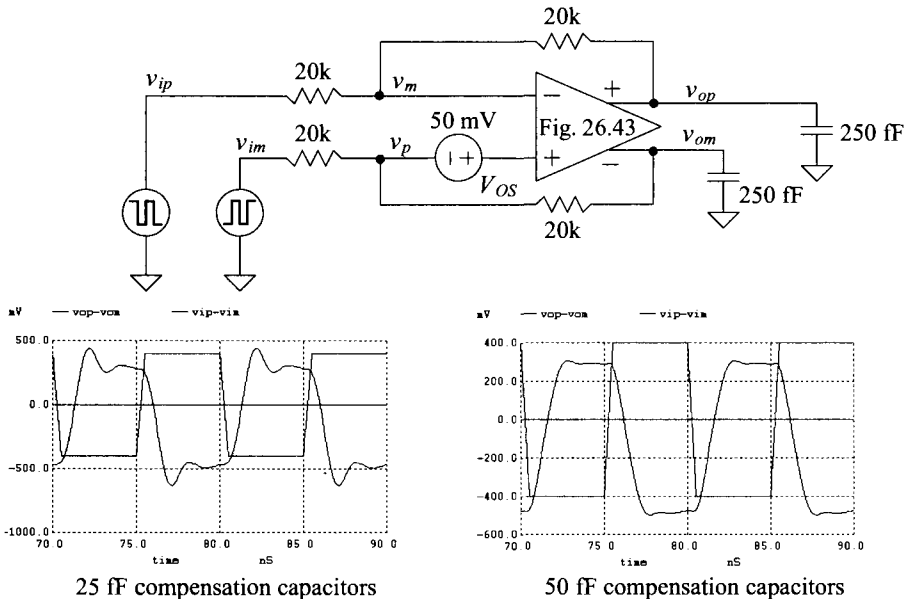


Figure 26.45 How an offset can affect the step response (compensation).

Op-Amp Offset Effects on Outputs

Notice, in Fig. 26.45, that an offset shifts the differential output signal by twice the offset voltage. The individual op-amp output voltages remain centered around V_{CM} . To describe this in more detail, consider the test setup seen in Fig. 26.46. We know the CMFB forces

$$\frac{v_{op} + v_{om}}{2} = V_{CM} \text{ or } v_{op} = 2V_{CM} - v_{om} \quad (26.8)$$

Further, assuming large op-amp open-loop gain,

$$v_p + V_{OS} = v_{pos} \approx v_m \quad (26.9)$$

Equating currents, we can write

$$\frac{V_{CMI} - v_p}{R_{in}} = \frac{v_p - v_{om}}{R_f} \text{ or } \frac{V_{CMI} - v_m + V_{OS}}{R_{in}} = \frac{v_m - V_{OS} - v_{om}}{R_f} \quad (26.10)$$

and

$$\frac{V_{CMI} - v_m}{R_{in}} = \frac{v_m - v_{op}}{R_f} \quad (26.11)$$

Subtracting Eq. (26.11) from Eq. (26.10), we get

$$\frac{V_{OS}}{R_{in}} = \frac{-V_{OS} + v_{op} - v_{om}}{R_f} \quad (26.12)$$

or

$$v_{op} - v_{om} = V_{OS} \cdot \left(\frac{R_f}{R_{in}} + 1 \right) \quad (26.13)$$

The voltages on the outputs of the op-amp are then

$$v_{op} = \frac{V_{OS}}{2} \cdot \left(1 + \frac{R_f}{R_{in}} \right) + V_{CM} \text{ and } v_{om} = V_{CM} - \frac{V_{OS}}{2} \cdot \left(1 + \frac{R_f}{R_{in}} \right) \quad (26.14)$$

The op-amp's input voltages are

$$v_m \approx v_p + V_{OS} = \frac{V_{OS}}{2} + V_{CM} \cdot \left(\frac{R_{in}}{R_{in} + R_f} \right) + V_{CMI} \cdot \left(\frac{R_f}{R_{in} + R_f} \right) \quad (26.15)$$

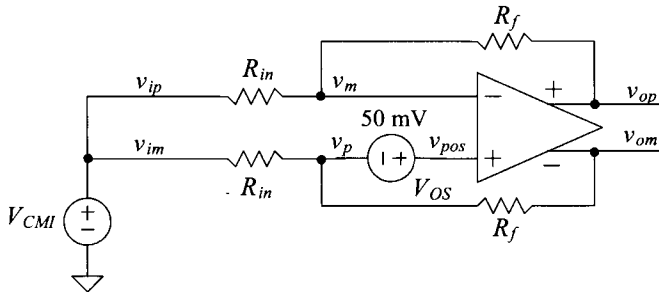


Figure 26.46 How an offset voltage causes an imbalance in the outputs.

If the input signal's common-mode voltage, V_{CMI} , is the same as the op-amp's common-mode voltage, V_{CM} , and V_{OS} is zero, then both op-amp inputs are held at V_{CM} . If $V_{CM} = V_{CMI}$ but the offset isn't zero, then $v_m = V_{OS}/2 + V_{CM}$ and $v_p = V_{CM} - V_{OS}/2$. Neglecting the offset voltage, if the input signal's common mode voltage isn't V_{CM} , then the op-amp's input common-mode voltage ($v_p = v_m$) will be different from the ideal value of V_{CM} . This can shut the op-amp off and cause undesirable behavior.

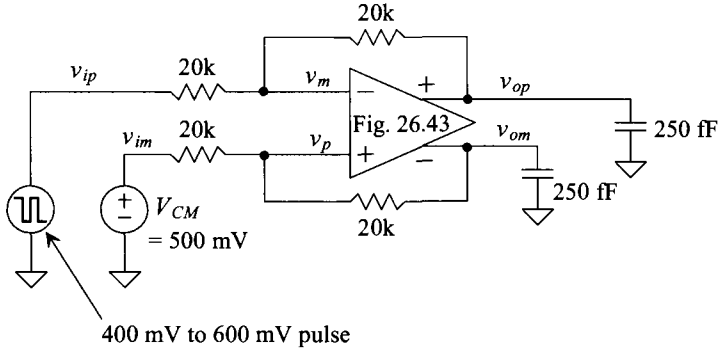


Figure 26.47 Problems with single-ended input signals.

Single-Ended to Differential Conversion

Even if the input signals are referenced around V_{CM} and the op-amp is offset free, we can still have problems. Consider the circuit in Fig. 26.47. In this circuit the input signal is single-ended. The other input to the op-amp is tied to V_{CM} . Even though the input is balanced around 500 mV, it is not a truly differential signal. When the input signal is up at 600 mV, the effective input common-mode voltage, V_{CMI} , is 550 mV (the average of the two inputs). When the input signal is at 400 mV, V_{CMI} is 450 mV. Figure 26.48 shows the simulation results using the circuit in Fig. 26.47. These results are very interesting. Consider what's happening between 70 and 75 ns. During this time, v_{ip} is 600 mV and v_{im} is 500 mV (and so $v_{ip} - v_{im} = 100$ mV). The op-amp's inputs move above the ideal 500 mV, as seen in the figure. This has the effect of increasing the transconductance of the diff-amp (the diff-amp's tail current is operating near/in the triode region so the tail

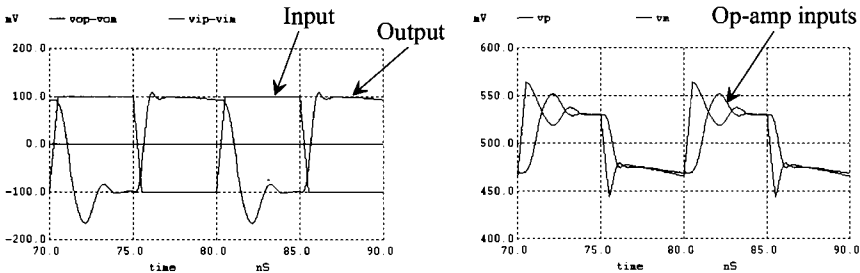


Figure 26.48 Simulating the operation of the circuit in Fig. 26.47.

current increases as the input common-mode voltage goes up). The increase in the diff-amp's g_m causes the unity gain frequency to increase and, with a fixed compensation capacitor, the phase margin to decrease (causing overshoot and ringing in the circuit's step response). Next consider what happens between 75 and 80 ns. The input signal switches to $400\text{ mV} - 500\text{ mV} = -100\text{ mV}$ and the op-amp's inputs drop down to around 475 mV. This causes the diff-amp's g_m to decrease and the op-amp to slow down (become more stable). However, at such a low input common-mode voltage the gain of the op-amp will drop and the outputs may wander. (See Sec. 30.3.1 for design information).

CMFB Settling Time

If we look at the voltage, V_{CMFB} , in the op-amps in Fig. 26.40 or 26.43, we may see that they aren't settling as quickly as the differential mode signals. Is this bad? As seen in Eq. (26.13), variations in V_{CM} don't (ideally) affect the differential output signal. However, if the variations in V_{CMFB} and thus V_{CM} are large, one of the output signals can saturate at close to V_{DD} or ground (the op-amp's gain drops), affecting the differential signal path.

CMFB in the Output Buffer (Fig. 26.43) or the Diff-Amp (Fig. 26.40)?

In this section we've presented the design of op-amps using continuous-time CMFB. Various design trade-offs and topologies were presented. At this point a good question is: "Which CMFB scheme is better?" Controlling the output common-mode level through the output buffer (Fig. 26.43) is simple, easy to ensure stability, and fairly robust. However, consider what happens if the op-amp in Fig. 26.43 is used in the topology in Fig. 26.46 (without an offset). If V_{CM} is ground, then, according to Eq. (26.15), the input voltages to the op-amp will move to 250 mV while the outputs of the op-amp remain at 500 mV. Thinking about this for a moment and neglecting the fact that an op-amp input common-mode voltage of 250 mV will shut the op-amp off, we see that the outputs of the op-amp must source a current back through the feedback resistors to the inputs. Further, anytime $V_{CM} < V_{CM}$, the PMOS in the output buffer must source a DC current back to the inputs. Similarly, if $V_{CM} > V_{CM}$, the NMOS in the output buffer must sink a current from the inputs of the op-amp. When using CMFB in the output buffers, we have no way of increasing the quiescent or DC current flowing in the output buffer to source/sink current from the input source (or to a DC load connected to ground). The output voltage of the diff-amp in the op-amp of Fig. 26.43 biases, or sets, the quiescent current in the output buffer. The CMFB scheme used in the output buffer simply adjusts the drive strength of the NMOS (in the output buffer) to set the op-amp's output common-mode level.

Using the CMFB scheme in Fig. 26.40, we can adjust the output voltage of the diff-amp (and ultimately the op-amp's output voltage) and thus the bias current in the output buffer (both the NMOS and the PMOS connected to the output buffer). However, if the output buffer must source/sink a significant amount of current, we can have problems with this topology too. For example, if the diff-amp's output voltage increases (turning off the PMOS in the output buffer and attempting to pull the op-amp's outputs down) while the NMOS devices connected to the outputs are sinking significant current, then it's possible that the outputs will get pulled upwards and the CMFB will fail (likely causing the gain of the op-amp to drop and the outputs to have limited swing).

To ensure the most robust op-amp design (biasing tolerant to offsets and easy to compensate the CMFB loops), CMFB circuits can be placed around both op-amp stages. This method is used in our last op-amp design example discussed next.

26.4 Op-Amp Design Using Switched-Capacitor CMFB

In this section we turn our attention towards op-amp designs for switched-capacitor (SC) circuits. We'll develop an op-amp design based on the topologies discussed in the last section. In this section, however, we'll use SC CMFB instead of continuous-time CMFB. Again, we'll use the nanometer CMOS process with minimum lengths and (roughly) 100 mV overdrive voltages (for drain currents of 20 μ A, see Fig. 26.18). We'll stick with two-stage designs because of the low open-circuit gains present when doing high-speed design.

Clock Signals

The SPICE listing for the clock signals used in the simulations in this section is seen below. Note that the 100 MHz phi1 and phi2 clocks are used with the NMOS transistors (phi1 and phi2 are not high at the same times), while the complements of these clock signals are used with the PMOS switches (not low at the same times).

```
*Clock Signals
Vphi1 phi1 0 DC 0 Pulse 0 1 0 200p 200p 4n 10n
Vphi1b phi1b 0 DC 0 Pulse 1 0 0 200p 200p 4n 10n
Vphi2 phi2 0 DC 0 Pulse 0 1 5n 200p 200p 4n 10n
Vphi2b phi2b 0 DC 0 Pulse 1 0 5n 200p 200p 4n 10n
R1 phi1 0 1MEG
R2 phi1b 0 1MEG
R3 phi2 0 1MEG
R4 phi2b 0 1MEG
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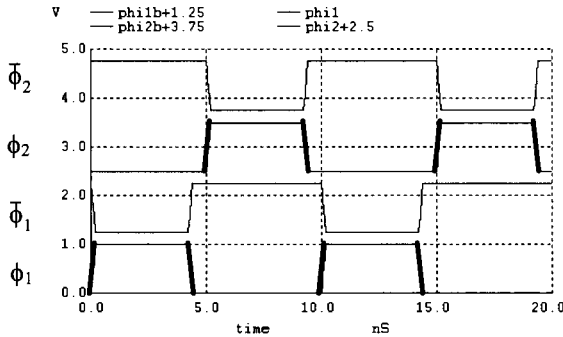


Figure 26.49 Generating nonoverlapping clocks for SC circuits.

Switched-Capacitor CMFB

Figure 26.50 shows the SC CMFB circuit from Fig. 26.16 along with a symbolic representation. We've selected 10 fF capacitors for the high-speed averaging capacitors. The kT/C noise associated with these capacitors is a common-mode signal and so it shouldn't affect the differential-mode signal. We don't want these capacitors to be too large because they will load the output of the amplifier. We made the switched-capacitors associated with the differencing and averaging 50 fF so that the changes in V_{CMFB} during one clock cycle won't be too large due to differences between the ideal V_{CMFB} and the actual V_{CMFB} . When the average of the outputs, v_{op} and v_{om} , is above their ideal value, the

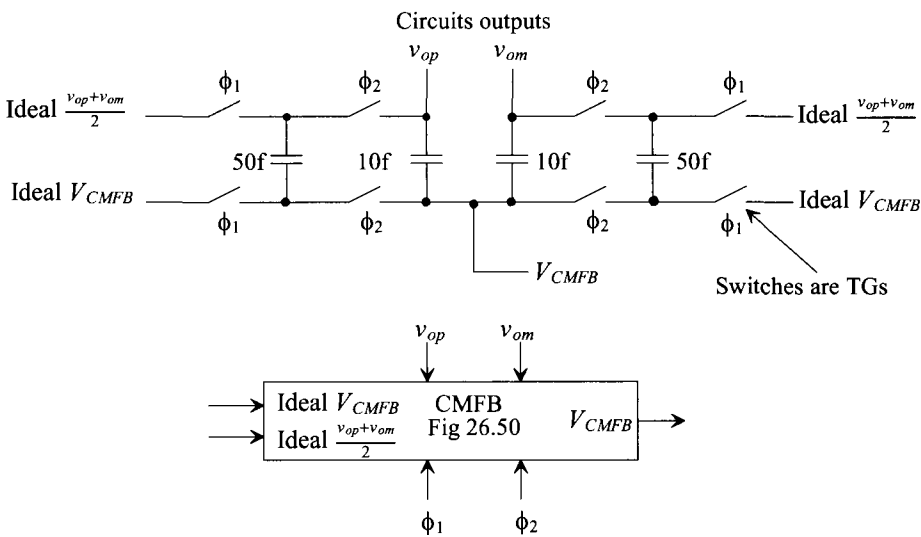


Figure 26.50 A switched-capacitor CMFB circuit (see Fig. 26.16). Switches implemented with transmission gates.

CMFB circuit's outputs moves upwards. It's important that this causes the average value of the outputs to move downwards. In other words, we've got to ensure that negative feedback is used in the CMFB loop.

Figure 26.51 shows some simulation results using the SC CMFB in Fig. 26.50. The ideal V_{CMFB} is 400 mV. The ideal average output, $\frac{v_{op}+v_{om}}{2}$, is 500 mV (V_{CM}). Prior to 150 ns, both v_{op} and v_{om} are 500 mV. At 150 ns, these two voltages jump (in the simulation) to 700 mV (they jump 200 mV away from the ideal average output of 500 mV). The voltage, V_{CMFB} , also jumps at this time. The key point to notice is that V_{CMFB} moves in the same direction as do changes in the output common-mode voltage. Knowing this is important when we are ensuring that our CMFB loop employs negative feedback.

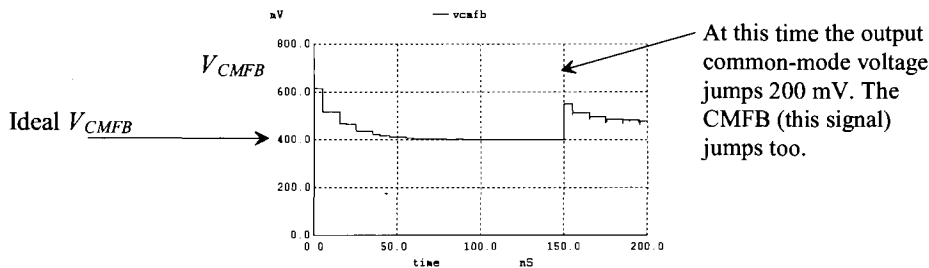


Figure 26.51 How the common-mode feedback signal increases to pull the outputs down.

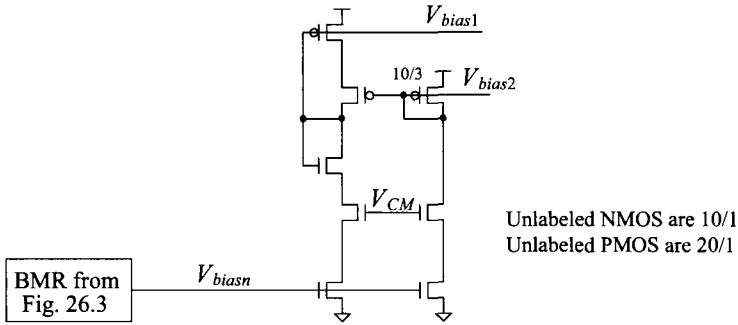


Figure 26.52 Biasing circuit for the op-amp developed in this section.

The Op-Amp's First Stage

As just mentioned, the output of the SC CMFB circuit, V_{CMFB} , moves in the same direction as movement in the average of the amplifier's outputs, v_{op} and v_{om} (the output common-mode level). Keeping this in mind, consider the bias circuit and diff-amp schematics seen in Figs. 26.52 and 26.53. We've separated the bias circuit out from the diff-amp, see Fig. 26.43, for a couple of reasons. To begin, an increase in the CMFB signal in Fig. 26.43 caused the outputs of the diff-amp to increase. As just mentioned, we want the diff-amp's outputs to move in the opposite direction of V_{CMFB} . Moving the CMFB-controlled MOSFET into the tail current of the diff-amp provides this control. Next, our bias circuit may now be shared with several op-amps. The benefit of sharing the bias circuit is a reduction in power dissipation.

Before adding the CMFB circuit, let's look at a SC circuit using an op-amp. Reviewing the sample and hold in Fig. 26.25, we see that when the ϕ switches are closed, the op-amp's inputs and outputs are at (ideally) V_{CM} . The op-amp, during this time, is in

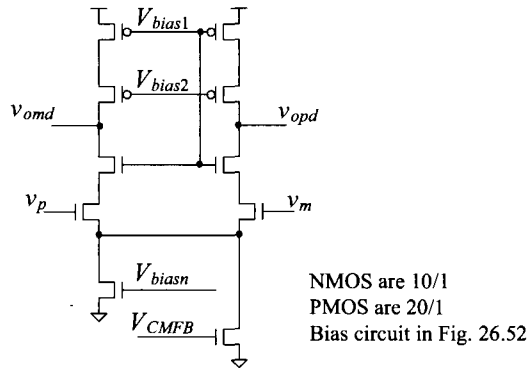


Figure 26.53 Diff-amp used with the bias circuit of Fig. 26.52.

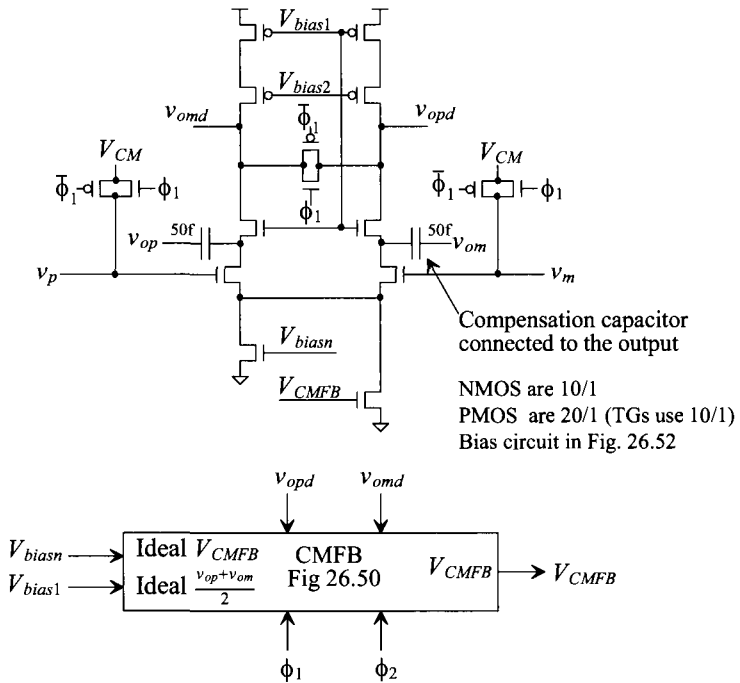


Figure 26.54 First-stage diff-amp with SC CMFB.

the unity-follower configuration. Instead of placing the op-amp in the follower configuration during this time, let's: 1) short the inputs to op-amp (Fig. 26.54) to V_{CM} , 2) short the outputs of the diff-amp (the inputs to the output buffer) together, 3) use a SC CMFB circuit around the diff-amp to ensure a balance condition (and to set the diff-amp's output voltage to V_{bias1}), 4) short the outputs of the op-amp together (so now both the output buffers inputs are shorted and outputs are shorted), and 5) use a SC CMFB around the output buffer to ensure balanced outputs. In other words, during ϕ_1 , the differential inputs are shorted together. At the same time, the outputs of the op-amp are shorted together.

Figure 26.55 shows the simulated output of the diff-amp in Fig. 26.54. The SC CMFB circuit sets the outputs of the diff-amp at V_{bias1} or roughly 600 mV. By shorting the two diff-amp outputs together and connecting the inputs to V_{CM} , we are ensuring that even with horrible mismatch, the outputs of the diff-amp will be equal and can be used to bias the output buffer. If, for example, the diff-pair shows a 50 mV mismatch (V_{OS}) then, when ϕ_1 is high, the drain currents in each side of diff-amp are set differently (making this scheme very tolerant to offsets). When ϕ_1 goes low and the op-amp is placed in a feedback configuration, each of the diff-amp's inputs will move (in opposite directions) by $V_{OS}/2$ (and so this topology doesn't store the op-amp's offset voltage).

Note that an important concern, as discussed in the last section, is the input common-mode voltage of the diff-amp. If, for example, the input common-mode voltage